WORKED SOLUTION

# STEP II 1999, Question 4

Pure • Binomial Coefficient Identities

**STEP II 1999, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q4

## The key idea

When two polynomials are identically equal, their coefficients of each power of $x$ agree. So we pick a product of two binomial expansions whose $x^n$ coefficient is the sum we want, and then compute the same coefficient from a simpler form of the product. The only real choice is which two factors to multiply: changing $x$ to $-x$ in one factor brings in the signs for (i), and differentiating one factor brings in the extra $t$ for (ii).

## Solution

### The first result

By the binomial theorem,

$$(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r.$$

Multiply two copies of this together. A term in $x^n$ comes from taking $x^s$ from the first bracket and $x^{n-s}$ from the second, for some $s$ with $0 \le s \le n$. So the coefficient of $x^n$ in $(1+x)^n(1+x)^n$ is

$$\sum_{s=0}^{n} \binom{n}{s}\binom{n}{n-s} = \sum_{s=0}^{n} \binom{n}{s}^2,$$

using the symmetry $\binom{n}{n-s} = \binom{n}{s}$, which is clear from the factorial formula. On the other side of the identity, the coefficient of $x^n$ in $(1+x)^{2n}$ is $\binom{2n}{n}$. The two polynomials are identical, so their $x^n$ coefficients are equal:

$$\sum_{s=0}^{n} \binom{n}{s}^2 = \binom{2n}{n}.$$

**Part (i)**

We need alternating signs, so we replace the first factor by $(1-x)^n = \sum_r \binom{n}{r}(-1)^r x^r$ and use the identity

$$(1-x)^n(1+x)^n \equiv (1-x^2)^n.$$

As before, the $x^n$ terms on the left come from $x^s$ in the first bracket and $x^{n-s}$ in the second, so the coefficient of $x^n$ is

$$\sum_{s=0}^{n}(-1)^s\binom{n}{s}\binom{n}{n-s} = \sum_{s=0}^{n}(-1)^s\binom{n}{s}^2.$$

On the right,

$$(1-x^2)^n = \sum_{k=0}^{n}\binom{n}{k}(-1)^k x^{2k}$$

contains only even powers of $x$. Since $n$ is even, $x^n$ does appear, namely when $2k=n$, that is $k=n/2$, and its coefficient is $(-1)^{n/2}\binom{n}{n/2}$. Equating coefficients,

$$\sum_{s=0}^{n}(-1)^s\binom{n}{s}^2 = (-1)^{n/2}\binom{n}{n/2},$$

as required. (The same argument shows the sum is 0 when $n$ is odd, because then $(1-x^2)^n$ has no $x^n$ term.)

**Part (ii)**

**Ask yourself** How can we make a factor of $t$ appear next to $\binom{n}{t}$?

Differentiation does it: $\dfrac{\mathrm{d}}{\mathrm{d}x}x^t = t\,x^{t-1}$. Differentiating the binomial expansion gives

$$n(1+x)^{n-1} = \sum_{t=1}^{n} t\binom{n}{t}x^{t-1}.$$

Now multiply by $(1+x)^n = \sum_u \binom{n}{u}x^u$ and use the identity

$$n(1+x)^{n-1}(1+x)^n \equiv n(1+x)^{2n-1}.$$

On the left, a term in $x^{n-1}$ comes from $t\binom{n}{t}x^{t-1}$ in the first bracket times $\binom{n}{n-t}x^{n-t}$ in the second, so the coefficient of $x^{n-1}$ is

$$\sum_{t=1}^{n} t\binom{n}{t}\binom{n}{n-t} = \sum_{t=1}^{n} t\binom{n}{t}^2.$$

On the right, the coefficient of $x^{n-1}$ in $n(1+x)^{2n-1}$ is $n\binom{2n-1}{n-1}$. Finally,

$$\binom{2n}{n} = \frac{(2n)!}{n!\,n!} = \frac{2n}{n}\cdot\frac{(2n-1)!}{(n-1)!\,n!} = 2\binom{2n-1}{n-1},$$

so equating coefficients and doubling,

$$\sum_{t=1}^{n} 2t\binom{n}{t}^2 = 2n\binom{2n-1}{n-1} = n\binom{2n}{n}.$$

*Otherwise:* there is a neat symmetry argument. Replacing $t$ by $n-t$ and using $\binom{n}{n-t} = \binom{n}{t}$ shows $\sum_{t=0}^{n} t\binom{n}{t}^2 = \sum_{t=0}^{n}(n-t)\binom{n}{t}^2$. Adding these two expressions for the same sum gives

$$\sum_{t=0}^{n} 2t\binom{n}{t}^2 = n\sum_{t=0}^{n}\binom{n}{t}^2 = n\binom{2n}{n}$$

by the first result, and the $t = 0$ term is zero.

## Checks

Take $n = 4$, where the rows of Pascal's triangle are $1, 4, 6, 4, 1$. Then $1 + 16 + 36 + 16 + 1 = 70 = \binom{8}{4}$; the alternating sum is $1 - 16 + 36 - 16 + 1 = 6 = (-1)^2\binom{4}{2}$; and $2(1 \cdot 16 + 2 \cdot 36 + 3 \cdot 16 + 4 \cdot 1) = 280 = 4 \times 70$. We also checked all three identities exactly for every $n$ up to 39.

## Common slips

- Forgetting that in part (i) the right-hand side has only even powers, so the evenness of $n$ is exactly what makes $x^n$ appear.
- Comparing the wrong power in part (ii): after differentiating, the exponents drop by one, so we need $x^{n-1}$, not $x^n$.

## Beyond STEP

The first identity is a special case of Vandermonde's identity, $\sum_s \binom{m}{s}\binom{n}{k-s} = \binom{m+n}{k}$, which also has a counting proof: choose $k$ people from $m$ women and $n$ men by deciding how many women to take. Using polynomials as bookkeeping devices for sequences is the method of *generating functions*, a central tool in combinatorics and probability. Grimmett and Welsh, *Probability: An Introduction*, uses generating functions throughout, and Siklos, *Advanced Problems in Mathematics*, has more questions in this style.

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